# Daily Number Theory Drill #1 — Speed Sheet

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| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Core Divisibility · Parity Arguments · Alternating Sums.*

## Section A Rapid Recognition

**1.** Factorise completely into primes: 2025.

Ans: $3^4 \times 5^2$

**2.** What is the remainder when $10^6$ is divided by 7?

Ans: 1

**3.** Find the last digit of $3^{2024}$.

Ans: 1

**4.** For what single digit $d$ is the number $73d4$ divisible by 9?

Ans: 4

**5.** The positive integers $a$ and $b$ are distinct. Given $\gcd(a, b) = 6$ and $\text{lcm}(a, b) = 36$, find $a + b$.

Ans: 42

**6.** Find the largest integer $k$ such that $2^k$ divides 20!.

Ans: 18

**7.** Find the remainder when $11^{10}$ is divided by 10.

Ans: 1

**8.** Find the remainder when $2^{100} + 3^{100}$ is divided by 5.

Ans: 2

**9.** What is the smallest prime factor of 1001?

Ans: 7

**10.** Find the only positive integer $n$ such that $n^2 - 1$ is a prime number.

Ans: 2

## Section B Manipulation Drills

**1.** Find the number of positive divisors of 360.

Ans: 24

**2.** A positive integer $x$ leaves a remainder of 3 when divided by 5, and a remainder of 5 when divided by 7. Find the smallest possible value of $x$.

Ans: 33

**3.** What is the sum of the digits of the number $2^{2024} \times 5^{2025}$?

Ans: 5

**4.** Solve for the smallest positive integer $x$ such that $4x \equiv 5 \pmod{7}$.

Ans: 3

**5.** Find the last two digits of $99^2$.

Ans: 01

**6.** Find the greatest common divisor of $2^{12} - 1$ and $2^{18} - 1$.

Ans: 63

**7.** The sum of three consecutive positive integers is a perfect cube. Find the smallest possible value for the largest of these three integers.

Ans: 10

**8.** Find the smallest positive integer $n$ such that $10n$ is a perfect square and $6n$ is a perfect cube.

Ans: 36000

**9.** Find the number of trailing zeros in 100!.

Ans: 24

**10.** The 5-digit number $24x8y$ is divisible by 4 and by 9. Find the maximum possible value of $x + y$.

Ans: 13

## Section C Substitution & Structure

**1.** Find the sum of all integers $x$ such that $\dfrac{x^2+5}{x+1}$ is an integer.

Ans: −8

**2.** Find the number of ordered pairs of positive integers $(a, b)$ such that $\dfrac{1}{a} + \dfrac{1}{b} = \dfrac{1}{6}$.

Ans: 9

**3.** For how many integers $n \in [1, 100]$ is the polynomial $x^n - 1$ divisible by $x^2 + x + 1$?

Ans: 33

**4.** The 5-digit number $A34B2$, where $A$ and $B$ are single digits and $A \neq 0$, is a multiple of 99. Find the value of the product $A \times B$. *(after SMC 2018 Q16)*

Ans: 18

**5.** The product of 24 integers is equal to 1. Their sum is 0. How many of the integers are equal to $-1$? *(after SMC 2015 Q11)*

Ans: 12

**6.** Find the largest integer $x$ such that $2^x$ divides $3^{256} - 1$.

Ans: 10

**7.** Find all pairs of primes $(p, q)$ such that $p^2 - 2q^2 = 1$.

Ans: (3, 2)

**8.** Find all prime numbers $p$ such that $p^2 + 2$ is also prime. You may first wish to show that if $p > 3$ then $p^2 - 1$ is divisible by 24. *(after TMUA 2017 Paper 2 Q9 and TMUA 2022 Paper 2 Q3)*

Ans: 3

## Section D Challenge

**1.** Find the sum of all integers $n$ such that $\dfrac{n^3 + 100}{n + 10}$ is an integer.

Ans: −510

**2.** How many 5-digit numbers can be formed using only the digits 1, 2, and 3, such that the number is a multiple of 3?

Ans: 81

**3.** Let $P(x)$ be a polynomial with integer coefficients such that $P(1) = 10$ and $P(5) = 2$. Find all possible integer roots of $P(x)$.

Ans: 6

**4.** Find the number of positive divisors of the smallest positive integer $n$ such that $\frac{n}{2}$ is a perfect square, $\frac{n}{3}$ is a perfect cube, and $\frac{n}{5}$ is a perfect fifth power.

Ans: 1232

**5.** Find all pairs of positive integers $(x, y)$ such that $x^2 - y! = 2016$.

Ans: (84, 7)